

Regulation functions by counting states

Four promoters of increasing architecture, and **one procedure applied to all four**, with less of my working shown each time. **We do items 1 and 2 in the room** — nine minutes, and item 1 is already done. Items 3 and 4 are on **PS3**.

The procedure, every time — the same four boxes as the slides:

1	List every state the promoter can be in.	3	Say what each state fires at.
2	Give each state a weight.	4	$\text{rate} = \alpha \frac{\sum(\text{weight} \times \text{fires-at})}{\sum \text{weight}}$

Throughout: $a = [A]/K_A$ is an occupancy variable, p is the polymerase weight, and $f = e^{-\epsilon_{ap}/k_B T}$ is charged **only** to a state in which two proteins are actually touching. **Rates are in units of α** , the rate a promoter fires at with polymerase bound and nothing else — so that column reads 0, 1 or f , and α multiplies the whole expression and never changes. Every answer below is taken in the **weak-promoter limit** ($p \ll 1$), where the fold-change is the regulation factor and p cancels.

1 • Fully worked — simple repression at a truncated lac promoter

lacUV5 cut down to a single operator O_m , with LacI₄ binding it. The operator **overlaps the polymerase site**: the repressor and RNAP cannot both be bound.

List the states. Three, not four. The missing one is what makes this repression.

STATE	WEIGHT	FIRES AT ($\times \alpha$)
empty	1	0
LacI ₄ on O_m	$r = [R]/K_m$	0
RNAP on the promoter	p	1
<i>both</i>	—	this state cannot exist

Rate-weight the rows and divide by the total. Only one row fires, so the sum on top has one term:

$$\frac{\text{rate}}{\alpha} = \frac{0 \cdot 1 + 0 \cdot r + 1 \cdot p}{1 + r + p} = \frac{p}{1 + r + p}$$

Take the ratio to the unregulated promoter ($r = 0$), which is what an experiment reports:

$$\text{fold-change} = \frac{p/(1 + r + p)}{p/(1 + p)} = \frac{1 + p}{1 + r + p}$$

Take the weak-promoter limit, $p \ll 1$:

$$F_{\text{reg}} = \frac{1}{1 + [R]/K_m}$$

which is exactly the form you derived in session 4 from a binding equilibrium — reached here by counting states, with no polymerase weight left to cancel.

► **Why does that step follow?** There is no f anywhere in this answer. Answer from the *mechanism*, not the algebra: what would f have been charged for, and what about this architecture means there is nothing to charge?

2 · The last step is yours — cooperative activation at λP_{RM}

Two operators. cl_2 at O_{R2} **activates** — it contacts polymerase, with enhancement factor f . cl_2 at O_{R1} does **not** contact polymerase at all; it helps recruit the other dimer, through a cooperative interaction ω between the two bound dimers. One protein species, so both occupancies move together: write $a = [cI_2]/K_{R2}$ and $h = [cI_2]/K_{R1}$.

List the states, and give each a weight. Four of them. Two are filled in.

Read the convention before you fill it in, because it is not the one on the slides. This table is already in the weak-promoter limit: the polymerase has been divided out, so every row here has RNAP bound and the states listed are the states of **cl only**. "No cl" therefore fires at 1, not at 0 — it is the unregulated promoter, which is what everything else is measured against.

STATE OF THE OPERATORS	WEIGHT	FIRES AT ($\times \alpha$)
no cl bound	1	1
cl_2 on O_{R2} only	a	f
cl_2 on O_{R1} only		
both		

► Where does ω go, and where does it not? One line, before you write the answer down.

Now divide, and take the weak-promoter limit.

$$F_{\text{reg}} =$$

Two checks, and do both — they are worth more than the derivation.

► Set $\omega = 1$, then separately set $h = 0$. One leaves the helper bound but not helping; the other deletes the helper site altogether. Do both, and then answer the question the result raises: **why do two different edits give the same expression, and what does that tell you about what ω is for?**

► Let $[cI_2] \rightarrow \infty$ — remembering that a and h both rise with it. Show that the answer is f , **whatever ω and K_{R1} are**. One line, and it is the whole design result: the helper site cannot raise the ceiling.

3 · On PS3 — two different proteins at *melAB*

MeIR*₂ binds the weak proximal operator *O2* and activates. CRP*₂ binds the upstream operator *O1*, helps recruit MeIR, and **does not itself activate**.

Two species now, so the two occupancies move independently. Write the state list, the regulation factor, and then answer the question that matters: **whose *f* appears in the saturating fold-change, and why is there only one of them?**

4 · On PS3 — the design item

Design a promoter whose fold-change is **AND-like** in two inducers: low unless both are present. Give the architecture, the state list, the regulation factor, and the truth table it produces.

Then say **which knob you turned** to make it AND-like rather than OR-like, and what you would have to measure to know whether your promoter actually does it.